

UrbanCab Fare Analysis

An end-to-end data analytics project analyzing urban cab rides to understand fare patterns, ride demand, vehicle performance, revenue, and supply-demand trends — combining **Python**, **SQL**, **PostgreSQL**, and **Power BI** to transform raw ride data into meaningful business insights.

DATA ANALYTICS PROJECT

POWER BI DASHBOARD



Made with GAMMA



Project Overview

UrbanCab ride data contains information about trips, fares, vehicle types, ride requests, driver availability, traffic levels, and demand-supply conditions. The project answers key business questions:

Revenue & Fare

Overall revenue, average fare, and which vehicle types generate the most revenue

Demand Patterns

When ride demand is highest and how demand compares between weekdays and weekends

Distance & Fare

How distance affects the final fare

City Activity

Which cities have the highest ride activity and which periods show higher demand-supply pressure

Tools & Technologies

The project combines Python, SQL, PostgreSQL, and Power BI to transform raw ride data into meaningful business insights.



Python

Data cleaning & exploratory data analysis



Pandas & NumPy

Data manipulation & numerical analysis



Matplotlib & Seaborn

Data & statistical visualization



PostgreSQL & SQL

Database analysis & business queries



Power BI

Interactive dashboard & business reporting



Jupyter Notebook

Python analysis environment

Dataset

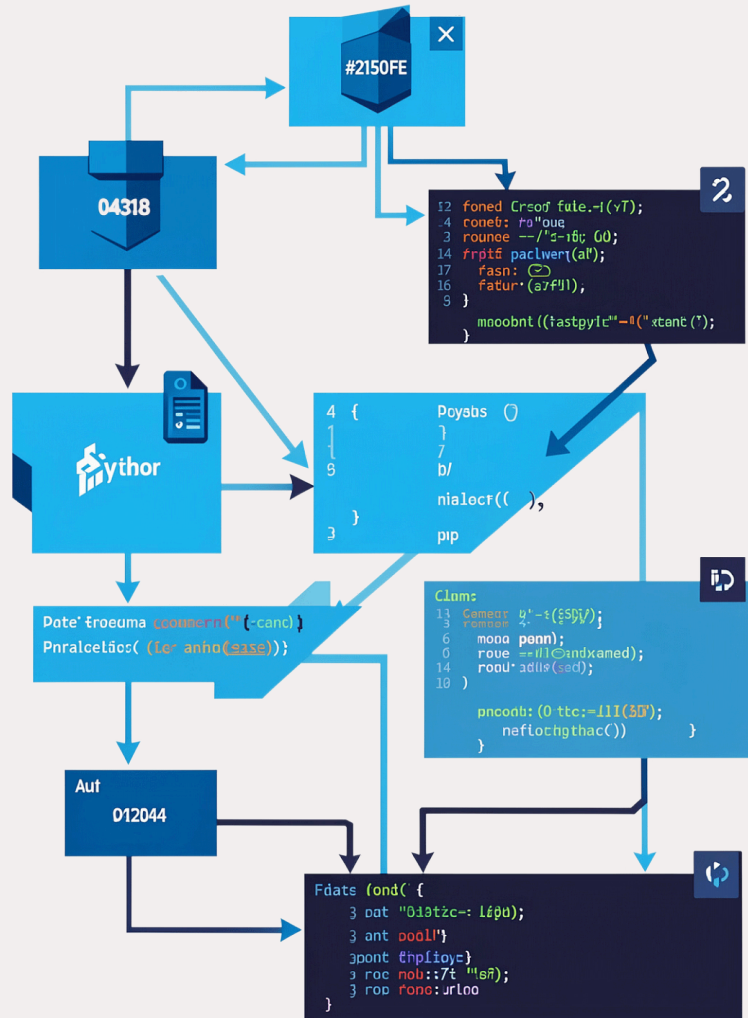
	UbON	MecODDS	Urban ab RRide	Roolo	#C4Drds
1	#215FE	\$29.07	7,50	630	#5,710
2	#215FE	\$2,800	6.40	\$90	#52D7
3	U4U5FE	\$2,501	8,60	8.00	#6,80
3	#431S	\$2,500	6.60	510	#32.D47
4	U451C	\$2,300	\$,50	5.00	#4,80
5	Urban Cab	#2,800	1170	# 20	#19C4
7	Cb Ride	\$3,00	8.00	2.80	#10
5	#2150FF	\$2,100	33.00	5.00	#3,20
4	0431FF	\$3,500	8,80	5.10	#5,00
6	#2150D	\$6,500	1.60	780	#12.8F
3	#2394	\$5,500	5.00	\$20	#4.C4
5	#21DFF	\$3,000	1.90	\$.10	#C4C5

The dataset contains **1,000 urban cab ride records** with fields including:

- User_ID, Driver_Name, City, Ride_Date
- Hour_of_Day, Day_of_Week, Distance_km
- Traffic_Level, Type_of_vehicle
- No_of_active_drivers, Ride_Requests
- Demand_Supply_Ratio, Trip_Duration
- Base_Fare, Surge_Multiplier, Final_Fare

Additional fields created during analysis: **Time Period** and **Day Type**.

Project Workflow



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1. Data Understanding

Loaded dataset with Pandas, examined rows, columns, data types, descriptive statistics, missing values, duplicates, and numerical distributions

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2. Data Cleaning

Handled missing driver names, converted and validated data types, checked duplicates, investigated potential numerical outliers using the IQR method, and created additional analytical features

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3. Exploratory Data Analysis

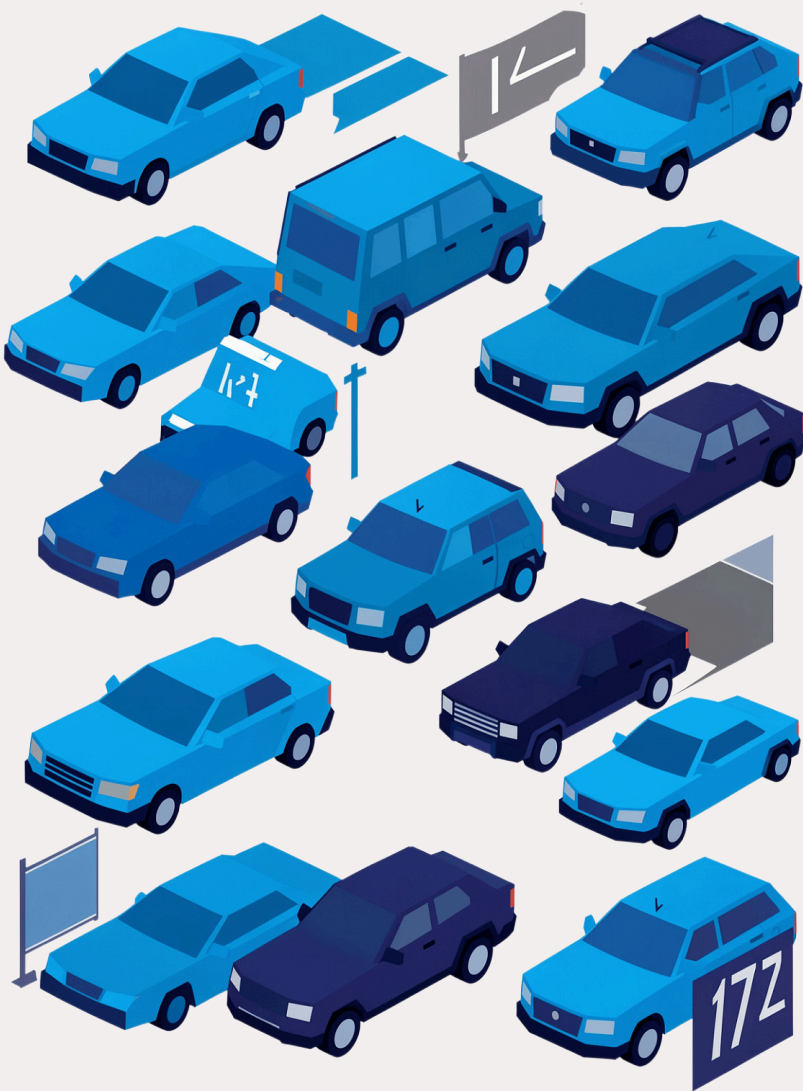
Analyzed fare distribution, distance vs final fare, revenue by vehicle type, ride demand by hour and day, driver availability, demand-supply ratio, and city-level ride activity

Fare Analysis

Fare Relationship

Analyzed the relationship between **Distance** → **Base Fare** → **Surge Multiplier** → **Final Fare**, examining fare distribution, distance vs final fare, average fare by vehicle type, fare per KM, and fare categories.





Vehicle Performance Analysis

Vehicle types were compared using multiple perspectives instead of evaluating performance using only revenue.

Total Revenue

Revenue generated per vehicle type

Average Fare

Average fare per vehicle type

Ride Volume

Ride count per vehicle type

Revenue Per Ride

Revenue per ride by vehicle type

Demand & Operations Analysis



Analyzed ride demand across:

→ **Hour of the Day**

Ride demand by hour

→ **Day of the Week**

Demand by day-of-week

→ **Weekday vs Weekend**

Weekday vs weekend demand comparison

→ **Peak Periods**

Periods where demand and available supply may differ

→ **Driver Availability**

Driver availability analysis

→ **Demand-Supply Ratio**

Demand-supply ratio analysis

SQL Analysis & Power BI Dashboard

SQL Analysis

PostgreSQL was used to perform business-oriented SQL analysis, including:

- Total revenue & average fare
- Revenue by vehicle type
- Peak-hour ride demand
- Top cities by ride count
- Average fare by vehicle
- Revenue by fare category
- City revenue ranking

Queries used SELECT, WHERE, GROUP BY, ORDER BY, aggregate functions, CASE statements, and CTEs.

Power BI Dashboard

The Power BI dashboard converts the analysis into an interactive business-reporting solution.

Demand & Operations: Hourly ride demand, day-of-week demand, peak periods, demand-supply analysis, driver availability

Fare & Customer Analysis: Total revenue, average fare, fare distribution, distance vs fare, fare per KM, vehicle-wise fare analysis

Features: KPI cards, interactive charts, slicers, trend analysis, category comparisons, business-focused visual storytelling

Business Value & Key Skills

This project demonstrates how raw transportation data can be transformed into actionable analytical insights:

Raw Data	Python EDA	SQL Analysis
Power BI Dashboard	Business Insights	

Skills demonstrated: Data Cleaning, EDA, Data Validation, Outlier Analysis, Feature Engineering, Pandas, NumPy, Matplotlib, Seaborn, SQL (SELECT, WHERE, GROUP BY, ORDER BY, Aggregates, CASE, CTEs), Power BI, KPI Development, Dashboard Design, Data Visualization, PostgreSQL.

— **Siril K S** | Aspiring Data Analyst | Python | SQL | Power BI | PostgreSQL